

Unit 2 – Veterinary Mycology

All 10 syllabus topics on four working sheets. Fungi are a small unit that carries disproportionate marks because the questions repeat: name the fungus from its microscopic picture, name the toxin from the lesion, name the medium. This sheet set is built around exactly those three reflexes.

PAPER	SYLLABUS TOPICS	REVISION TIME	COURSE
Paper I · Units 1, 2, 3	10 of 10 covered	20 min full pass	B.V.Sc & A.H. II Yr
Theory 100 · Practical 60		8 min sheet 04 only	VCI MSVE · 3+2=5

Sheet	What it covers	Syllabus topics folded in
01	The fungal cell, classification, growth & reproduction, lab toolkit	t01–t02
02	Yeasts & yeast-like fungi — <i>Candida</i> , <i>Cryptococcus</i> , <i>Malassezia</i> ; dermatophytes	t03, t05
03	Moulds, dimorphic fungi, zygomycetes, mycetoma	t04, t06–t08
04	Mycotic mastitis & abortion, mycotoxicoses, antifungals	t09–t10

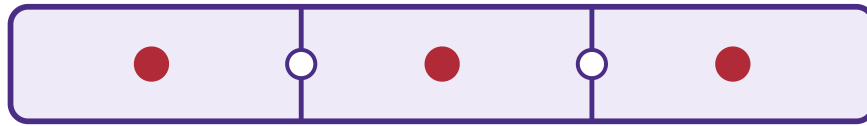
■ General mycology
 ■ Yeasts & dimorphic
 ■ Moulds
 ■ Dermatophytes
 ■ Mycotoxins

■ Exam trap & mnemonic

What a fungus is – and is not

MECHANISM

SEPTATE HYPHA



ASEPTATE / COENOCYTIC HYPHA



SEPTATE HYPHA

Septum with pore · nucleus — true, membrane-bound · cytoplasm continuous

ASEPTATE / COENOCYTIC HYPHA

Zygomycetes — *Mucor*, *Rhizopus*, *Absidia*.
Ribbon-like, wide, irregular

Wall = CHITIN + glucan + mannan

(bacteria = peptidoglycan) → β -lactams useless
Membrane sterol = **ERGOSTEROL** (mammals = cholesterol)

Eukaryote

True nucleus, 80S ribosome, mitochondria, ER, Golgi. Heterotroph, absorptive nutrition

Why this matters therapeutically: every antifungal in the exam targets one of the two things above — **ergosterol** (polyenes bind it, azoles block its synthesis, allylamines block squalene epoxidase) or **chitin/glucan** (echinocandins block β -1,3-glucan synthase). Griseofulvin is the exception: it binds tubulin and blocks mitosis.

Fungi vs bacteria

TABLE

Feature	Fungi	Bacteria
Cell type	Eukaryote	Prokaryote
Nucleus	True, membrane-bound	Nucleoid
Ribosome	80S	70S
Cell wall	Chitin, glucan	Peptidoglycan
Membrane sterol	Ergosterol	Absent (except <i>Mycoplasma</i>)
Size	2–10 μm and above	0.2–5 μm
Growth temp.	25–30 °C (room)	37 °C
Optimal pH	5.6 (acidic)	7.2–7.4
Growth speed	Days to weeks	Hours
Reproduction	Spores — sexual & asexual	Binary fission

Feature	Fungi	Bacteria
Oxygen	Aerobe / facultative	Full range

SDA Sabouraud Dextrose Agar — 4% dextrose, peptone, **pH 5.6**, incubated at **25–27 °C**. Selectivity is added with **chloramphenicol** (kills bacteria) and **cycloheximide/actidione** (suppresses saprophytic moulds). **DTM** (dermatophyte test medium) adds phenol red → turns **red** with a dermatophyte.

Classification you must be able to write

TABLE

Division	Hyphae & sexual spore	Veterinary examples
Zygomycota	Aseptate; zygospore	<i>Mucor</i> , <i>Rhizopus</i> , <i>Absidia</i> , <i>Basidiobolus</i> , <i>Conidiobolus</i>
Ascomycota	Septate; ascospore in an ascus	<i>Aspergillus</i> , <i>Penicillium</i> , <i>Histoplasma</i> , <i>Blastomyces</i> , <i>Microsporium</i> , <i>Trichophyton</i> , <i>Candida</i> (teleomorph)
Basidiomycota	Septate + clamp connections; basidiospore	<i>Cryptococcus neoformans</i> , mushrooms, rusts & smuts
Deuteromycota "Fungi Imperfecti"	Septate; no known sexual stage	Older home of most veterinary pathogens; now redistributed by molecular data

- **Anamorph** = asexual (imperfect) state · **Teleomorph** = sexual (perfect) state · **Holomorph** = the whole fungus.
- **Morphological classes:** **Yeast** (unicellular, buds, pasty colony) · **Yeast-like** (buds + pseudohyphae — *Candida*) · **Mould/filamentous** (hyphae + mycelium, cottony colony) · **Dimorphic** (mould at 25 °C, yeast at 37 °C in tissue).
- **Clinical classification of mycoses:** **Superficial** (*Malassezia*, piedra) → **Cutaneous** (dermatophytes) → **Subcutaneous** (sporotrichosis, mycetoma, rhinosporidiosis) → **Systemic/deep** (histoplasmosis, blastomycosis, coccidioidomycosis) → **Opportunistic** (candidiasis, aspergillosis, cryptococcosis, zygomycosis).
- **Mycotoxigenesis** and **mycetismus** (mushroom poisoning) are **intoxications, not infections** — no fungus grows in the animal.

Reproduction — spore names in one pass

RAPID-FIRE

Asexual — conidia	Borne on a conidiophore, exogenous, not enclosed
Microconidia / macroconidia	Small single-celled / large multiseptate — the basis of dermatophyte identification
Blastoconidia	Formed by budding — <i>Candida</i> , <i>Cryptococcus</i>
Arthroconidia	Hyphal fragmentation — <i>Coccidioides</i> , <i>Geotrichum</i>
Chlamydoconidia	Thick-walled resting cell; terminal chlamydospore on cornmeal-Tween 80 = <i>C. albicans</i>
Sporangiospore	Endogenous , inside a sporangium — Zygomycetes
Phialoconidia	From a flask-shaped phialide — <i>Aspergillus</i> , <i>Penicillium</i>

Sexual — zygospor	Fusion of two similar gametangia (Zygomycota)
Ascospore	Usually 8, inside a sac (ascus)
Basidiospore	4, on the outside of a club-shaped basidium
Oospore	Fusion of unequal gametangia (Oomycetes — <i>Pythium insidiosum</i>)

The distinction that gets asked every year: sporangiospores are **endogenous** (formed inside a sac), conidia are **exogenous** (formed free at the tip). That one line separates Zygomycetes from everything else.

Growth requirements

RAPID-FIRE

Temperature	25–30 °C for moulds; 37 °C selects the yeast phase of dimorphics
pH	Acidic, ~5.6
Humidity	High; $a_w > 0.70$ permits mould growth on feed
Nutrition	Heterotrophic, absorptive; secrete exoenzymes
Time	Report a fungal culture as negative only after 3–4 weeks
Oxygen	Aerobic; <i>Candida</i> facultatively anaerobic
Colony types	Glabrous (waxy), velvety, cottony/floccose, granular, powdery
Pigment	Note obverse and reverse colour — reverse pigment is a key ID clue

Diagnostic toolkit

RAPID-FIRE

10–20% KOH mount	Digests keratin, leaves fungal elements — skin, hair, nail
LPCB mount	Lactophenol cotton blue: lactic acid preserves, phenol kills, glycerol prevents drying, cotton blue stains chitin
India ink / nigrosin	Negative stain — <i>Cryptococcus</i> capsular halo in CSF
Slide culture	Riddell's technique — preserves conidial arrangement <i>in situ</i>
Germ tube test	Serum, 37 °C, 2–3 h — Reynolds-Braude phenomenon , <i>C. albicans</i>
Special stains	PAS (magenta), GMS Grocott methenamine silver (black), calcofluor white + UV, mucicarmine for <i>Cryptococcus</i>
Wood's lamp	365 nm UV — apple-green fluorescence , <i>Microsporum canis</i>
Serology / molecular	Latex agglutination for cryptococcal antigen, galactomannan & β-D-glucan for aspergillosis, AGID, ELISA, ITS-region PCR & sequencing , MALDI-TOF

Sheet 01 • 60-second self-test

RECALL

Prompt	Answer
Fungal cell wall polymer	Chitin (+ glucan, mannan)

Prompt	Answer
Fungal membrane sterol	Ergosterol
pH and temperature of SDA	5.6 · 25–27 °C
Antibacterial added to SDA	Chloramphenicol
Inhibitor of saprophytic moulds	Cycloheximide (actidione)
Aseptate hyphae division	Zygomycota
Endogenous asexual spore	Sporangiospore
Dye in LPCB that stains chitin	Cotton blue
Slide culture technique	Riddell's
Germ tube phenomenon	Reynolds-Braude

02 Yeasts, Yeast-like Fungi & Dermatophytes

t03, t05

Candida albicans

SYSTEMATIC

- **Morphology:** Gram-positive oval budding yeast 2–6 µm; forms **blastoconidia**, **pseudohyphae** and **true hyphae** — the only common yeast that makes all three. Normal flora of gut, mouth, genital tract → a true **opportunistic**.
- **Culture:** SDA 25–37 °C, 24–48 h — creamy-white, pasty, **yeasty smell**. On **CHROMagar**, *C. albicans* = green, *C. tropicalis* = blue, *C. krusei* = pink rough.
- **Two confirmatory tests:** **Germ tube test** — serum at 37 °C for 2–3 h gives a filament with **no constriction at its base** (constricted base = pseudohypha of another species). **Cornmeal-Tween 80 agar** at 25 °C → **thick-walled terminal chlamydospores**.
- **Predisposing causes:** prolonged broad-spectrum antibiotics, corticosteroids, immunosuppression, indwelling catheters, high-carbohydrate diet, moist skin folds.
- **Veterinary disease: crop mycosis / thrush (sour crop) in poultry** — thickened crop with a white pseudomembrane and "Turkish-towel" appearance; oral thrush and gastric candidiasis in young pigs and calves; mycotic mastitis in cows after intramammary infusion; vaginitis, cystitis and otitis in dogs.
- **Virulence:** adhesins, **phenotypic switching**, hyphal invasion, secreted aspartyl proteinases (SAPs), phospholipase, biofilm on catheters.
- **Treatment:** nystatin (topical/oral, not absorbed), fluconazole, amphotericin B; copper sulphate in poultry water; correct the predisposing cause.

Cryptococcus neoformans

SYSTEMATIC

- **Morphology:** spherical budding yeast 4–20 µm with a **huge polysaccharide capsule (glucuronoxylomannan)** and a **narrow-based single bud**. **No pseudohyphae**.
- **Demonstration: India ink / nigrosin negative stain of CSF** — clear halo against a dark background; **mucicarmine** stains the capsule rose-red; latex agglutination for capsular antigen is the sensitive test.

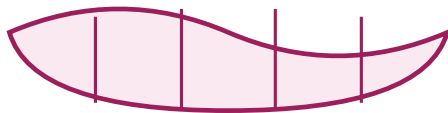
- **Culture:** SDA **without cycloheximide** (it is inhibited by it) — mucoid cream colonies. **Urease positive**. **Birdseed / niger seed agar** → brown-black colonies from **phenol oxidase making melanin**.
- **Source:** **pigeon (and other avian) droppings** and soil; *C. gattii* from eucalyptus. Birds are carriers, not cases — their body temperature is too high.
- **Disease:** inhalation → **nasal granuloma with a "Roman nose" swelling in cats** (the classic case), sinusitis, pneumonia, **meningoencephalitis** with gelatinous cystic "soap-bubble" lesions in brain; mastitis in cattle; disseminated disease in dogs and horses. Zoonotically important in immunosuppressed people, but transmission is environmental, not animal-to-man.
- **Virulence:** capsule (antiphagocytic, the single most important factor), **melanin**, phospholipase, growth at 37 °C, urease.
- **Treatment:** amphotericin B + flucytosine; fluconazole penetrates CSF well; itraconazole for nasal forms.

Yeast in tissue — decide in one step: capsule and narrow-based bud = *Cryptococcus* · pseudohyphae = *Candida* · **broad-based** bud = *Blastomyces* · **multiple buds, "mariner's wheel"** = *Paracoccidioides* · intracellular in macrophages = *Histoplasma* · spherule with endospores = *Coccidioides*.

Dermatophytes — identify by macroconidium

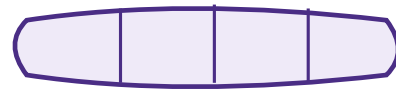
MECHANISM

MICROSPORUM



Large, spindle-shaped, rough/echinulate thick wall,

TRICHOPHYTON



MICROSPORUM

Large, spindle-shaped, **rough/echinulate** thick wall, 6–15 septa. Macroconidia abundant, microconidia few.

M. canis — dog & cat, **Wood's lamp +ve**, reverse yellow

M. gypseum — geophilic, soil, horses; *M. nanum* — pigs

TRICHOPHYTON

Cigar / pencil-shaped, **smooth thin** wall, 2–6 septa. **Microconidia abundant** (grape-like clusters, above).

T. verrucosum — **cattle ringworm**, needs thiamine + inositol, grows best at 37 °C. *T. mentagrophytes* — rodents, dogs, horses

EPIDERMOPHYTON

Club-shaped smooth macroconidia in twos and threes, **NO microconidia**; skin & nail only, never hair; human pathogen

Ringworm — the answer in eight lines

SYSTEMATIC

- **Definition:** dermatophytosis = keratinophilic infection of the **non-living keratinised tissue** — stratum corneum, hair and nail. The fungus never invades living tissue; the lesion is largely the host's inflammatory response.
- **Ecology:** **anthropophilic** (man) · **zoophilic** (animals — *M. canis*, *T. verrucosum*) · **geophilic** (soil — *M. gypseum*). **Zoophilic strains cause the most severe inflammation in man** — this is the standard zoonosis question.

- **Hair invasion pattern: Ectothrix** — arthroconidia sheathing the outside of the shaft (*Microsporum*, *T. verrucosum*) · **Endothrix** — inside the shaft, hair breaks at scalp level · **Favic** — hyphae + air spaces (*T. schoenleinii*, scutula).
- **Lesion:** circular, expanding, alopecic, scaly, grey-white crust with **central healing and an active periphery** — the "ring". In calves, classic on head, neck and around the eyes; in horses under the girth and saddle.
- **Diagnosis:** pluck hairs from the **edge** of the lesion → **KOH mount** for ecto/endothrix → **Wood's lamp** (only ~50% of *M. canis* fluoresce; *Trichophyton* does not) → SDA + chloramphenicol + cycloheximide at 25 °C for up to 4 weeks → LPCB for macroconidia → DTM turns red → PCR.
- **Treatment:** topical enilconazole, miconazole, 2% lime sulphur, povidone-iodine; systemic **griseofulvin** (with fatty food, **teratogenic — never in pregnancy**), itraconazole, terbinafine. Clip and burn hair, disinfect premises with hypochlorite; spores survive in the environment for years.
- **Prophylaxis:** live attenuated *T. verrucosum* vaccine (LTF-130) is used in cattle in some countries.

Malassezia pachydermatis

SYSTEMATIC

- **Morphology:** small oval-to-bottle-shaped budding yeast with a characteristic **broad-based "footprint" / peanut or bowling-pin shape** and a **monopolar unipolar bud with a collarette**.
- **Key point:** *M. pachydermatis* is **non-lipid-dependent** and therefore **grows on plain SDA** — unlike the human species (*M. furfur*) which need an olive-oil overlay.
- **Disease:** normal skin and ear-canal flora of dogs that overgrows with allergy, endocrinopathy, moisture or excess ceruminous secretion → **ceruminous otitis externa and seborrheic dermatitis of dogs**, greasy rancid smell, erythema and lichenification of skin folds. Also intertrigo and paronychia.
- **Diagnosis:** tape-strip or ear swab, stained with Diff-Quik — counting >5 yeasts per oil-immersion field supports the diagnosis. Culture is confirmatory.
- **Treatment:** miconazole-chlorhexidine shampoo, clotrimazole ear drops, oral ketoconazole/itraconazole, plus correction of the underlying allergy.

Other superficial Piedra — hair-shaft nodules: black piedra (*Piedraia hortae*, hard) and white piedra (*Trichosporon*, soft). **Pityriasis versicolor** — *M. furfur*, "spaghetti and meatballs" on KOH.

Sheet 02 · 60-second self-test

RECALL

Prompt	Answer
Two confirmatory tests for <i>C. albicans</i>	Germ tube · chlamydospore on cornmeal agar
Sour crop / thrush in poultry	<i>Candida albicans</i>
Negative stain for <i>Cryptococcus</i>	India ink / nigrosin
Birdseed agar detects	Phenol oxidase → melanin, <i>Cryptococcus</i>
Environmental source of <i>Cryptococcus</i>	Pigeon droppings
Rough, spindle-shaped macroconidia	<i>Microsporum</i>
Smooth, cigar-shaped macroconidia	<i>Trichophyton</i>

Prompt	Answer
Cattle ringworm agent	<i>T. verrucosum</i> (thiamine + inositol)
Wood's lamp positive dermatophyte	<i>M. canis</i> — apple-green
Antifungal contraindicated in pregnancy	Griseofulvin
Yeast of canine otitis externa	<i>Malassezia pachydermatis</i>

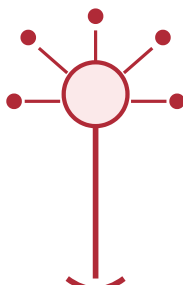
03 Moulds, Dimorphic Fungi & Subcutaneous Mycoses

t04, t06–t08

Aspergillus vs *Penicillium* vs *Zygomycetes*

MECHANISM

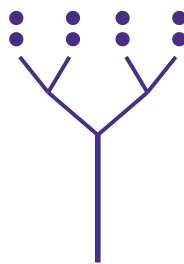
ASPERGILLUS



ASPERGILLUS

Foot cell + swollen **vesicle** + phialides radiating in a whole or half circle

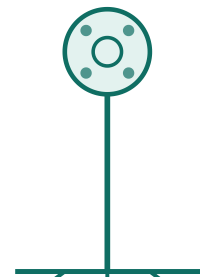
PENICILLIUM



PENICILLIUM

No vesicle — branched brush ("penicillus"), chains of conidia

RHIZOPUS



RHIZOPUS

Sporangium + columella; rhizoids **opposite** the stolon = *Rhizopus*

Zygomycete separation: *Rhizopus* — rhizoids **directly opposite** the sporangiophore · *Absidia* — rhizoids **between** two sporangiophores, pear-shaped sporangium with apophysis · *Mucor* — **no rhizoids at all**. All three are **angio-invasive**, causing thrombosis and infarction, and all grow fast, filling the plate in 2–3 days ("lid-lifters").

Aspergillosis & penicilliosis

SYSTEMATIC

- *A. fumigatus* — the chief pathogen; blue-green velvety colony, **uniseriate phialides on the upper half of the vesicle only, thermotolerant (grows at 45 °C)**. *A. flavus* — yellow-green, biseriate, radiate head, **aflatoxin**. *A. niger* — black, biseriate, radiate, ochratoxin, "black fungus" otomycosis. *A. terreus*, *A. nidulans* less common.
- **Brooder pneumonia of chicks** — the classic vet lesion: inhalation of spores from mouldy litter or hatchery contamination; **yellow-white caseous nodules/plaques in lung and air sacs**, gasping, high mortality in the first 2 weeks. Also **mycotic keratitis, encephalitis and ophthalmitis** in birds.
- **Other hosts: mycotic abortion in cattle** (raised, leathery placentitis with cotyledonary necrosis and ringworm-like plaques on the foetal skin); **guttural pouch mycosis** in horses (fatal epistaxis from internal carotid erosion); nasal aspergillosis in dogs (**turbinate destruction, *A. fumigatus***); air-sacculitis and disseminated disease in penguins and raptors.

- **Virulence:** small hydrophobic conidia (2–3 µm) reaching the alveoli, thermotolerance, gliotoxin, elastase, catalase, adhesion to fibrinogen and laminin.
- **Diagnosis:** KOH/LPCB shows **dichotomously branching septate hyphae at 45°**; culture on SDA **without cycloheximide**; histopathology with GMS/PAS; **galactomannan and β-D-glucan** assays; AGID; PCR. **Positive culture alone is not proof** — *Aspergillus* is a ubiquitous contaminant; correlate with lesions.
- **Penicillium** — mostly a contaminant and a toxin producer (ochratoxin A, citrinin, patulin, PR toxin); *Talaromyces (Penicillium) marneffei* is the one dimorphic species, causing disseminated disease in bamboo rats and man in South-East Asia.

Dimorphic fungi – the "25 °C / 37 °C" table

TABLE

Fungus	Mould form, 25 °C	Tissue/yeast form, 37 °C	Disease
<i>Histoplasma capsulatum</i>	Tuberculate macroconidia (thick-walled with finger-like projections) + smooth microconidia	Small intracellular yeast inside macrophages , 2–4 µm	Histoplasmosis — reticulo-endothelial, chronic wasting & diarrhoea in dogs; bird & bat guano source. var. farciminosum = epizootic lymphangitis of horses ("Japanese farcy") — corded lymphatics with ulcerating nodules
<i>Blastomyces dermatitidis</i>	Small round-to-pear-shaped conidia on short stalks ("lollipops")	Large 8–15 µm yeast, thick doubly-contoured wall, single BROAD-BASED bud	Blastomycosis — pyogranulomatous pneumonia, skin nodules, uveitis, bone lesions in dogs; river valleys, moist soil
<i>Coccidioides immitis</i>	Barrel-shaped arthroconidia alternating with empty cells — highly infectious, BSL-3	Spherule (sporangium) 20–200 µm packed with endospores — not a yeast	Coccidioidomycosis / "valley fever" — desert soil, dust-borne; granulomatous pneumonia and disseminated bone disease in dogs and horses
<i>Paracoccidioides brasiliensis</i>	Slow-growing, sparse conidia	Yeast with multiple narrow-necked buds — "mariner's wheel / ship's wheel"	South American blastomycosis; man; armadillo reservoir

Dimorphism Mould in nature and on SDA at 25 °C; **yeast (or spherule) in the host at 37 °C**. The switch is driven by **temperature** (and CO₂/cysteine for *Histoplasma*), and the **mould form is the infectious form** while the **yeast form is the invasive form** — that sentence answers the standard "define dimorphism with examples" question.

Subcutaneous mycoses

SYSTEMATIC

- *Sporothrix schenckii* — **sporotrichosis**. Dimorphic; 25 °C mould with **daisy/rosette clusters of conidia on delicate hyphae**; 37 °C **cigar-shaped yeast**. Entry by thorn or splinter → "**rose-gardener's disease**"; in horses and cats **ascending chains of nodules and ulcers along the**

lymphatics (lymphocutaneous form). **Cats shed enormous numbers of organisms — a genuine zoonotic risk to handlers.** Asteroid bodies in tissue. Treat with itraconazole or oral potassium iodide.

- ***Rhinosporidium seeberi* — rhinosporidiosis. Cannot be cultured;** now placed with the aquatic protistan class Mesomycetozoea, not true fungi. Associated with stagnant pond water and river bathing. **Pedunculated, friable, red polyp studded with white spots ("strawberry polyp")** in the nose of cattle, horses, dogs and man; also conjunctiva. Histology shows **huge sporangia (up to 300 µm) full of endospores.** Treatment is **surgical excision with cauterization of the base** — recurrence is common; dapson is adjunctive.
- **Mycetoma (Madura foot).** A **triad**: tumefaction (swelling) + draining sinuses + **grains/granules** in the discharge. **Eumycotic** mycetoma = true fungi (*Madurella*, *Pseudallescheria boydii*, *Curvularia*) with **broad septate hyphae** in the grain; **Actinomycotic** mycetoma = filamentous bacteria (*Nocardia*, *Actinomyces*, *Streptomyces*) with **thin 1 µm filaments.** Grain colour and the Gram/ZN reaction of the grain decide which — and this decides the treatment: antifungals versus sulphonamides.
- **Zygomycosis (mucormycosis).** Opportunistic, in diabetic, acidotic, immunosuppressed or heavily antibiotic-treated animals. **Broad, aseptate, ribbon-like hyphae branching at wide (~90°) angles,** strongly **angio-invasive** → thrombosis, infarction and black necrotic tissue. Rhino-cerebral, gastric (calves, pigs) and abortion forms. Contrast with *Aspergillus*: septate, narrow, 45° branching.
- **Also name: pythiosis** ("swamp cancer" — *Pythium insidiosum*, an oomycete with kunkers, in horses and dogs) and **phaeohyphomycosis** (dematiaceous, melanised hyphae).

Sheet 03 • 60-second self-test

RECALL

Foot cell + vesicle	<i>Aspergillus</i>	Brush-like conidiophore, no vesicle	<i>Penicillium</i>
Brooder pneumonia	<i>A. fumigatus</i>	Guttural pouch mycosis	<i>A. fumigatus</i>
Rhizoids opposite the stolon	<i>Rhizopus</i>	No rhizoids	<i>Mucor</i>
Septate, 45° dichotomous branching	<i>Aspergillus</i>	Aseptate, 90° ribbon-like	Zygomycetes
Tuberculate macroconidia	<i>Histoplasma capsulatum</i>	Epizootic lymphangitis	<i>H. capsulatum</i> var. <i>farciminosum</i>
Broad-based single bud	<i>Blastomyces dermatitidis</i>	Spherule with endospores	<i>Coccidioides immitis</i>
Mariner's wheel	<i>Paracoccidioides brasiliensis</i>	Rosette/daisy conidia, cigar yeast	<i>Sporothrix schenckii</i>
Strawberry nasal polyp, non-cultivable	<i>Rhinosporidium seeberi</i>	Triad of swelling, sinuses, grains	Mycetoma

Mycotic mastitis

SYSTEMATIC

- **Agents:** *Candida* spp. (commonest), *Cryptococcus neoformans*, *Trichosporon*, *Aspergillus fumigatus*, *Geotrichum*, *Prototheca* (an achlorophyllous alga — always mentioned with this group).
- **How it happens — this is the whole answer:** it is nearly always **iatrogenic**. Repeated intramammary infusion of antibiotics with a contaminated or multi-dose tube, unhygienic teat canal, and the suppression of normal bacterial flora by prolonged antibiotic therapy together allow fungal overgrowth.
- **Clinical:** swollen, firm, painful quarter with fever in the acute phase, then a **self-limiting or chronic granulomatous mastitis that does not respond to antibiotics** — often the first clue. Milk is watery with flakes; marked drop in yield; the quarter may become fibrotic.
- **Diagnosis:** aseptic milk sample → direct smear (Gram/LPCB/Giemsa) showing budding yeasts or hyphae → SDA + chloramphenicol at 25 and 37 °C → germ tube/CHROMagar → confirm by repeat sampling, since a single positive can be contamination.
- **Management:** **stop all antibiotic infusions immediately**, frequent stripping, supportive therapy, nystatin or clotrimazole infusion, iodine teat dip, cull chronic cases. Control = strict aseptic infusion technique and single-dose tubes.

Prototheca zopfii is not a fungus but an **achlorophyllous alga**; it forms **morula-like sporangia with endospores**, grows on SDA, causes an incurable granulomatous mastitis (cull) and enteritis/disseminated disease in dogs.

Mycotic abortion

SYSTEMATIC

- **Agents:** *Aspergillus fumigatus* accounts for **60–80%**; also *Mucor*, *Rhizopus*, *Absidia*, *Candida*, *Mortierella wolfii* (New Zealand, with a fatal maternal pneumonia).
- **Pathogenesis:** inhalation or ingestion of spores from **mouldy hay, silage or damp bedding** in winter-housed cattle → **haematogenous spread** → localisation in the placentome → placentitis and foetal death. Not contagious cow-to-cow.
- **When:** sporadic, usually in the **last trimester (5–8 months)** in cattle; the dam is otherwise healthy.
- **Lesions — the two things to write:** **(1) Placenta** — severely thickened, leathery, necrotic cotyledons with a yellow-brown adherent exudate and oedematous intercotyledonary areas; retained. **(2) Foetus** — ~25% show **raised, grey-white, circumscribed "ringworm-like" plaques of dermatomycosis on the skin**, especially head, neck and shoulders — pathognomonic when present.
- **Diagnosis:** KOH/LPCB and GMS/PAS of cotyledon and foetal abomasal content showing **septate branching hyphae**; culture on SDA; histopathology showing hyphae invading placental vessels; **rule out *Brucella*, *Leptospira*, *Campylobacter*, IBR and BVD first** — say this in the answer.
- **Control:** no treatment. Prevent by avoiding mouldy feed, proper hay curing and silage compaction, ventilation, and mould inhibitors (propionic acid) in stored feed.

Mycotoxins – the single highest-yield table in this unit

TABLE

Mycotoxin	Producer	Substrate	Target organ & mechanism	Signs & the fact that gets asked
Aflatoxin B₁, B₂, G₁, G₂	<i>Aspergillus flavus</i> , <i>A. parasiticus</i>	Groundnut cake, maize, cottonseed	Liver — bioactivated by cytochrome P450 to the 8,9-epoxide which binds DNA	Most potent natural hepatocarcinogen . Ducklings & young poultry most susceptible; icterus, ascites, hepatic fibrosis, ↓ growth, immunosuppression. B₁ is excreted in milk as M₁ — the public-health point. Blue-green fluorescence at 365 nm; detected by TLC/HPLC/ELISA. Turkey X disease, 1960
Ochratoxin A	<i>A. ochraceus</i> , <i>Penicillium verrucosum</i>	Stored cereals, barley	Kidney — inhibits phenylalanine-tRNA synthetase	Nephropathy of pigs and poultry; pale swollen kidneys, polyuria, poor shell quality; also immunosuppressive and teratogenic
Fumonisin B₁	<i>Fusarium verticillioides</i> (<i>moniliforme</i>)	Maize	Blocks ceramide synthase → sphingolipid disruption	Equine leucoencephalomalacia (ELEM) — "mouldy corn poisoning", liquefactive necrosis of cerebral white matter; porcine pulmonary oedema ; hepatotoxic in other species
T-2 toxin & DON (vomitoxin)	<i>Fusarium sporotrichioides</i> , <i>F. graminearum</i>	Wheat, maize, damp grain	Trichothecene — inhibits protein synthesis at the ribosome; radiomimetic	Oral and gastric ulceration, beak & mouth lesions in poultry , feed refusal and vomiting in pigs (DON), bone-marrow suppression, haemorrhagic syndrome, contact dermatitis; alimentary toxic aleukia in man
Zearalenone (F-2)	<i>Fusarium graminearum</i>	Maize, barley	Oestrogen-receptor agonist	Hyperoestrogenism — vulvovaginitis, swollen vulva and prolapse in gilts, mammary enlargement, splay leg, infertility, anoestrus; the classic pig mycotoxicosis

Mycotoxin	Producer	Substrate	Target organ & mechanism	Signs & the fact that gets asked
Ergot alkaloids	<i>Claviceps purpurea</i> , <i>C. fusiformis</i>	Rye, bajra, grasses	α -adrenergic vasoconstriction; dopamine agonism → ↓ prolactin	Gangrenous ergotism — dry gangrene of extremities, ear tips, tail, hooves; convulsive form; agalactia in mares and sows. Fescue toxicosis (<i>Neotyphodium coenophialum</i> ergovaline) — fescue foot, summer slump, fat necrosis
Slaframine	<i>Rhizoctonia leguminicola</i>	Red clover ("black patch")	Cholinergic (muscarinic) agonist	Slobbers syndrome — profuse salivation, lacrimation, bloat
Sporidesmin	<i>Pithomyces chartarum</i>	Pasture litter (rye grass)	Bile-duct necrosis → phylloerythrin retention	Facial eczema of sheep — hepatogenous secondary photosensitisation of unpigmented skin
Citrinin · Patulin · PR toxin	<i>Penicillium</i> spp.	Rice, apples, silage	Nephrotoxic / cytotoxic	Yellow rice disease; often asked as a one-line item only

Diagnosis & control of any mycotoxicosis **Diagnosis:** herd history of a new feed batch + multiple animals affected + no fever + no response to antibiotics + non-contagious → examine feed for mould, UV screening, TLC/HPLC/LC-MS/MS, ELISA kits, and a **feeding trial**. **Control:** harvest and dry to **<14% moisture ($a_w < 0.70$)**, aerated storage, mould inhibitors (propionic/sorbic acid), **adsorbents** — **hydrated sodium calcium aluminosilicate (HSCAS)**, **bentonite**, **esterified glucomannan**, ammoniation for aflatoxin, dilution with clean feed, added antioxidants, methionine and vitamins. There is **no specific antidote** — remove the feed.

Antifungal drugs by target

TABLE

Class	Mechanism	Drugs & use
Polyenes	Bind ergosterol → membrane pores → leakage	Amphotericin B (systemic, nephrotoxic), nystatin (topical/oral only, not absorbed — candidiasis), natamycin
Azoles	Inhibit 14-α-demethylase (CYP51) → blocks ergosterol synthesis	Imidazoles — miconazole, ketoconazole, clotrimazole, enilconazole; Triazoles — fluconazole (best CSF penetration) , itraconazole, voriconazole
Allylamines	Inhibit squalene epoxidase	Terbinafine — dermatophytosis
Echinocandins	Inhibit β-1,3-glucan synthase (cell wall)	Caspofungin, micafungin — invasive candidiasis
Antimetabolite	Converted to 5-FU → blocks DNA/RNA	Flucytosine — always with amphotericin B for cryptococcosis
Others	Binds tubulin , blocks mitosis	Griseofulvin — dermatophytes only, teratogenic ; potassium iodide for sporotrichosis; copper sulphate & gentian violet as feed/water antifungals

Antifungal sensitivity testing CLSI broth microdilution (M27 for yeasts, M38 for moulds) giving an **MIC**; disc diffusion and **E-test** gradient strips are the practical lab versions. Less standardised than bacterial ABST — say so.

Sheet 04 • 60-second self-test

RECALL

Prompt	Answer
Commonest cause of mycotic mastitis	<i>Candida</i> spp.
Alga causing bovine mastitis	<i>Prototheca zopfii</i>
Commonest cause of mycotic abortion	<i>Aspergillus fumigatus</i>
Skin plaques on aborted foetus	Foetal dermatomycosis
Most potent natural hepatocarcinogen	Aflatoxin B ₁
Aflatoxin metabolite in milk	Aflatoxin M ₁
Turkey X disease, 1960	Aflatoxicosis
Equine leucoencephalomalacia toxin	Fumonisin B ₁
Hyperoestrogenism in gilts	Zearalenone
Vomitoxin	Deoxynivalenol (DON)
Facial eczema of sheep	Sporidesmin (<i>Pithomyces</i>)
Slobbers syndrome	Slaframine
Safe feed moisture level	Below 14%
Aflatoxin binder in feed	HSCAS / bentonite
Antifungal that binds ergosterol	Amphotericin B (polyene)
Antifungal binding tubulin	Griseofulvin

10 / 10 syllabus topics

Paper I • Unit 2

Full pass ≈ 20 min

Night-before: sheet 03 diagrams + all self-tests